# *Retail Store Sales & Inventory Performance Analysis*

## Project Overview:

This project focuses on analyzing retail store sales and inventory performance using Python and Power BI. The dataset contains 73,100 records with 15 original columns covering sales, products, inventory, pricing, discounts, and demand forecasts from 2022 to 2024. The analysis helps understand sales trends, revenue performance, and inventory management.

## Project Objectives:

- To analyse retail sales and revenue performance.
- To understand the impact of discounts, promotions, and seasonality on sales.
- To compare sales performance across categories and regions.
- To identify overstock and low-stock situations.
- To improve inventory planning through data-driven insights.

---

## Data Sources

- **Source Description and Timeline:** Google Dataset Search **and** 2022-2024.
- **Domain**: Retail / Sales Analytics

---

## Problem Statement

- Difficulty in maintaining optimal inventory levels.
- Identifying overstock and low-stock situations.
- Understanding sales and revenue trends across categories.
- Analyzing the impact of promotions, discounts, and seasonal changes.
- Improving demand forecasting and inventory planning.

---

## Attribute (Column /Features) Details:

| Attribute Name | Data Type | Description |
|---|---|---|
| **Date** | String | The date of the sales and inventory record |
| **Store ID** | String | Identifies the store |
| **Product ID** | String | Identifies the specific product. |
| **Category** | Categorical | Shows the category of the product |
| **Region** | String | Shows the geographical region of the store |
| **Inventory Level** | Numeric | Number of products available in stock. |
| **Units Sold** | Numeric | Number of units sold |
| **Units Ordered** | Numeric | Number of units ordered for restocking |
| **Demand Forecast** | Numeric | Estimated demand for the product |
| **Price** | Numeric | Selling price of the product |

| | | |
|---|---|---|
| **Discount** | Numeric | Discount percentage offered on the product |
| **Weather Condition** | String | Weather condition on that day. |
| **Holiday/Promotion** | Boolean | Indicates whether a holiday or special offer was active |
| **Competitor Pricing** | Numeric | Price of a similar product from competitors |
| **Seasonality** | String | Shows the season associated with the record. |

## Additional Columns Created for Data Analysis

| Attribute Name | Data Type | Description |
|---|---|---|
| Year | **String** | **Extracts the year from the date column.** |
| Month | String | Extracts the month from the date column. |
| Revenue | Numeric | Calculates total sales revenue before discounts. |
| Net Revenue | Numeric | Calculates revenue after deducting discounts. |
| Stock Status | String | Identifies whether stock is Normal, Overstock, or Low Stock. |
| Demand Difference | Numeric | Calculates the difference between forecasted demand and units sold. |
| Estimated Remaining Stock | Numeric | Estimates the stock remaining after deducting units sold. |
| Price Difference | Numeric | Calculates the difference between product price and competitor pricing. |

---

## Tools & Technologies

- **Google Colab** - Python coding environment.
- **Pandas & NumPy** - Data manipulation and numerical calculations.
- **Matplotlib & Seaborn** - Data visualization.
- **Power BI** - Interactive dashboard creation and business insights.

---

## Data Pre-Processing (Python – Google Colab)

### Tasks Performed:

- Data Loading: Imported the retail dataset using Pandas.
- Data Inspection: Checked dataset structure, columns, data types, and statistical summaries.
- Missing Value Check: Checked missing values; no missing values were found.
- Duplicate Check: Checked duplicate records; no duplicate records were found.
- Date Transformation: Converted the Date column into datetime format and extracted Year and Month.
- Feature Engineering: Created additional calculated columns to support sales and inventory analysis.

---

## Analysis and Visualizations

### Python

- Category-wise Revenue – Bar Chart
- Stock Status Distribution – Count Plot
- Units Sold Distribution – Histogram
- Seasonality-wise Net Revenue – Box Plot
- Demand Forecast vs Units Sold – Scatter Plot
- Correlation Heatmap
- Sales and Inventory Relationship – Pair Plot
- Revenue by Category and Region – Heatmap
- Year-wise Category Revenue Trend – Line Chart
- Stock Status vs Inventory Level – Violin Plot

### Power BI – Dashboard

- Total Revenue – Card
- Net Revenue – Card
- Total Units Sold – Card
- Category-wise Revenue – Bar Chart
- Monthly Revenue – Line Chart
- Stock Status – Donut Chart
- Interactive Filters and Slicers



## Insights & Conclusions

### Key Findings

- Furniture generated the highest revenue, while Electronics generated the lowest revenue.
- Overstock records are higher compared to Normal and Low Stock records.
- Most sales transactions fall within the lower units-sold range.
- Demand Forecast and Units Sold show a strong positive relationship.
- Revenue performance varies across product categories and regions.
- Groceries and Toys revenue decreased in 2023, while other categories showed slight growth.

---

## Analysis Insights

### A. Descriptive Analysis (What happened?)

- Total Net Revenue reached 494.97M.
- Total Units Sold reached 10M.
- Furniture generated the highest category-wise revenue.
- Overstock records were higher than other stock categories.

### B. Diagnostic Analysis (Why did it happen?)

- Demand Forecast and Units Sold have a strong positive correlation.
- Revenue varies across different product categories and regions.
- Weather conditions and promotions provide additional insights into sales variations.
- Inventory levels show differences across stock categories.

### C. Predictive Analysis (What may happen?)

- Demand Forecast helps estimate future product requirements.
- Products with increasing demand may require additional stock.
- Historical sales trends can help anticipate future sales patterns.
- Demand forecasting can support better inventory planning.

### D. Prescriptive Analysis (What should be done?)

- Reduce excess inventory to avoid unnecessary storage costs.
- Maintain sufficient stock for high-demand products.
- Improve stock replenishment based on demand forecasts.
- Introduce suitable promotions to improve sales.
- Monitor regional and seasonal sales performance to make better business decisions.

---

## Conclusions

This project successfully analysed retail sales and inventory data using Python and Power BI. It identified important sales trends, revenue performance, and stock management issues through data analysis and visualizations. The insights help businesses improve inventory planning, understand customer demand, and make better data-driven decisions.

---